

Scalability & Future Plan

Date	02 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document describes how the **Credit Card Approval Prediction** application can be enhanced to support a larger number of users, improve performance, and introduce new features in future versions. It also outlines the team's roadmap for extending the application's capabilities beyond its current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited integration with real-time banking and financial verification systems.	Restricts automated verification of applicant information.	High
2	Prediction history is stored in a local SQLite database.	Limits scalability for large-scale and multi-user environments.	High
3	Prediction accuracy depends on the quality and completeness of user input.	Incorrect or incomplete data may affect prediction reliability.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a moderate number of users on a single deployed instance.	Deploy using scalable cloud infrastructure with load balancing and auto-scaling.
2	Data Storage	Prediction history is maintained using SQLite.	Upgrade to PostgreSQL or MySQL for improved scalability and multi-user support.
3	Performance	Predictions are generated in real time for each request.	Optimize model inference, introduce caching, and improve backend performance.
4	Security	Basic input validation and secure deployment.	Implement user authentication, encrypted data storage, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User authentication, profile management, and cloud database integration.	2 Months	Improves security, data management, and user experience.
Phase 3	Explainable AI, advanced analytics, and model retraining using updated datasets.	4 Months	Enhances prediction transparency, accuracy, and overall system performance.
Phase 4	Mobile application, multilingual support, and integration with banking verification services.	6 Months	Improves accessibility, scalability, and enterprise-level usability.